

A RIGHT BREGMAN–CHEBYSHEV SET WITH NONCONVEX GRADIENT IMAGE

ABSTRACT. Let f be a Legendre function on $X = \mathbb{R}^n$, let $U = \text{int dom } f$, and let $C \subseteq U$. Bauschke, Macklem and Wang record that if $\text{dom } f = X$, if C is closed and nonempty with $\text{cl } \nabla f(C) \subseteq \text{int dom } f^*$, and if C is right D_f -Chebyshev, then $\nabla f(C)$ is convex, and they ask whether the hypothesis $\text{dom } f = X$ can be removed. It cannot. For f the negative entropy on $\mathbb{R}^2$ and $C = \{(e^t, e^{-t^2}) : 1 \leq t \leq 2\}$, the set C is compact and right D_f -Chebyshev at every point of U , the hypothesis $\text{cl } \nabla f(C) \subseteq \text{int dom } f^*$ holds, and $\nabla f(C)$ is a parabolic arc, hence nonconvex.

The same example applies to a theorem of Luo, Meng, Wen and Yao, who show that a variational condition (i) and the property (ii) that C is a right D_f -sun are equivalent when $U = X$. Our C is a right D_f -sun for which (i) fails, so $U = X$ cannot be removed from their theorem either. We also record, with attribution to the same authors, that the hypothesis $\text{cl } \nabla f(C) \subseteq \text{int dom } f^*$ is redundant.

1. INTRODUCTION

Let $X = \mathbb{R}^n$ with the standard inner product, and let $f: X \rightarrow]-\infty, +\infty]$ be proper, lower semicontinuous and convex. Write $U = \text{int dom } f$ and $U^* = \text{int dom } f^*$, where f^* is the Fenchel conjugate. Recall that f is *Legendre* if it is both essentially smooth and essentially strictly convex [1, 7]. For $x \in \text{dom } f$ and $y \in U$, the *Bregman distance* is

$$D_f(x, y) = f(x) - f(y) - \langle \nabla f(y), x - y \rangle.$$

It is not symmetric, so a set has two projection operators. Throughout, we use the *right* one, in which the second argument varies:

$$\overrightarrow{\Pi}_C(x) = \arg \min_{y \in C} D_f(x, y), \quad x \in U.$$

Following [3] we call C *right D_f -Chebyshev* if $\overrightarrow{\Pi}_C(x)$ is a singleton for every $x \in U$. We abbreviate

$$C^* := \nabla f(C).$$

The starting point of this note is the following statement, recorded as Fact 3.2 of the survey of Bauschke, Macklem and Wang [3].

Fact 1.1 ([3, Fact 3.2]). Suppose that

- (a) $\text{dom } f = X$;
- (b) $C \subseteq U$ is closed and nonempty, and $\text{cl } C^* \subseteq U^*$;
- (c) C is right D_f -Chebyshev.

Then C^* is convex.

The survey attaches an open problem to this fact, which we restate.

Problem 1.2 ([3, Problem 2]). Is the hypothesis $\text{dom } f = X$ necessary? Either exhibit a Legendre f with $\text{dom } f \neq X$ and a closed nonempty right D_f -Chebyshev $C \subseteq U$ with C^* nonconvex, or show that the conclusion survives without full domain.

Date: August 13, 2026.

This investigation was carried out with substantial assistance from automated systems, and we omit a conventional byline rather than overstate the named researchers' personal contributions. *Nikol Savova*, University of Oxford, nikol.p.savova@gmail.com; *Sihao Huang*, independent researcher, sihao.c.huang@gmail.com.

We settle Problem 1.2 in the first of the two directions.

Theorem 1.3. *Let $X = \mathbb{R}^2$ and let f be the negative entropy,*

$$f(x) = g(x_1) + g(x_2), \quad g(s) = \begin{cases} s \ln s - s, & s > 0, \\ 0, & s = 0, \\ +\infty, & s < 0. \end{cases}$$

Fix real numbers a, b with $1/\sqrt{2} \leq a < b$ and put

$$C = \{(e^t, e^{-t^2}) : t \in [a, b]\}.$$

Then f is Legendre with $\text{dom } f = \mathbb{R}_+^2 \neq X$, and:

- (i) *C is a nonempty compact subset of $U = \mathbb{R}_{++}^2$;*
- (ii) *$\text{cl } C^* = C^* \subseteq U^* = \mathbb{R}^2$, so hypothesis (b) of Fact 1.1 holds;*
- (iii) *C is right D_f -Chebyshev, so hypothesis (c) holds;*
- (iv) *$C^* = \{(t, -t^2) : t \in [a, b]\}$ is not convex.*

Consequently Fact 1.1 fails if hypothesis (a) is deleted and (b), (c) retained.

Both of the remaining hypotheses hold in this example. Here $U^* = \mathbb{R}^2$ and C^* is compact, so $\text{cl } C^* \subseteq U^*$; had that failed, the example would delete two hypotheses at once and would not isolate (a). Uniqueness of the projection holds at every point of U , including points arbitrarily close to ∂U and points far from C , and rests on the single inequality $4t^2 - 2 \geq 0$, valid for $t \geq 1/\sqrt{2}$. This is also the only constraint the argument places on the interval $[a, b]$. The proofs below use no numerical computation.

Section 3 applies the example to a theorem of Luo, Meng, Wen and Yao [5], Section 4 records what is known about hypothesis (b), and Section 5 lists open questions.

Earlier examples. Bauschke, Wang, Ye and Yuan [2, Example 7.5], reproduced as Example 3.3 of the survey [3], work with $h(u) = e^{u_1} + e^{u_2}$ and the segment $A = \{(\lambda, 2\lambda) : 0 \leq \lambda \leq 1\}$. Since the right projection for f corresponds to the left projection for f^* [5, Lemma 3.2], the curve $\nabla h(A) = \{(e^\lambda, e^{2\lambda})\}$ is a nonconvex right D_f -Chebyshev set for the negative entropy, and full domain fails there too. Its gradient image is A , which is convex, so the conclusion of Fact 1.1 holds and the example leaves Problem 1.2 untouched. Theorem 1.3 replaces the convex segment A by the nonconvex arc $\{(t, -t^2)\}$; the work lies in showing that Chebyshevness survives on all of U .

On the dual side, Theorem 1.3 states that the arc $\{(t, -t^2) : a \leq t \leq b\}$ is a compact nonconvex set that is left Chebyshev for the Bregman distance of h at every point of $\mathbb{R}^2$. Since $\text{dom } f = X$ holds if and only if f^* is supercoercive, this is also the statement that supercoercivity cannot be removed from the corresponding left-projection theorem.

We do not claim the first nonconvex right D_f -Chebyshev set for the negative entropy, which is [2, Example 7.5], nor the duality on which the construction rests. Laude, Ochs and Cremers [4] give nonconvex examples for entropy-like kernels, with single-valuedness on a neighbourhood under relative prox-regularity rather than at every point of U ; they record Problem 1.2 as open. Themelis and Wang [6] develop Bregman operators on their natural domains.

What Theorem 1.3 adds is a set that is right D_f -Chebyshev on all of U , with nonconvex gradient image, and with hypothesis (b) intact, so that (a) is the only hypothesis of Fact 1.1 that fails. We know of no earlier example with these three properties together.

Theorem 1.3 concerns the statement of Fact 1.1, not the individual instance. Convexity of C^* occurs for many f of non-full domain, [2, Example 7.5] among them; what fails is the implication once (a) is deleted. The theorem also says nothing about which weaker hypothesis might serve in place of (a), for which see Section 5.

2. THE EXAMPLE

Throughout this section f , C , a and b are as in Theorem 1.3, and we write

$$c(t) = (e^t, e^{-t^2}), \quad t \in [a, b],$$

so that $C = c([a, b])$. All closures are taken in $X = \mathbb{R}^2$.

Lemma 2.1. *The function f is Legendre, with*

$$\text{dom } f = \mathbb{R}_+^2, \quad U = \mathbb{R}_{++}^2, \quad \nabla f(x) = (\ln x_1, \ln x_2) \quad \text{on } U,$$

and

$$f^*(u) = e^{u_1} + e^{u_2}, \quad \text{dom } f^* = U^* = \mathbb{R}^2.$$

Moreover $\nabla f: U \rightarrow U^*$ is a bijection with inverse $u \mapsto (e^{u_1}, e^{u_2})$.

Proof. For $s > 0$ we have $g'(s) = \ln s$ and $g''(s) = 1/s > 0$, so g is strictly convex on $]0, \infty[$. Strict convexity persists at the left endpoint: for $b_0 > 0$ and $0 < \lambda < 1$,

$$g(\lambda b_0) = \lambda g(b_0) + \lambda b_0 \ln \lambda < \lambda g(b_0) + (1 - \lambda)g(0),$$

since $\ln \lambda < 0$ and $g(0) = 0$. As $\lim_{s \downarrow 0} (s \ln s - s) = 0 = g(0)$, the function g is lower semicontinuous on $\mathbb{R}$. Hence f is proper, lower semicontinuous and convex, with $\text{dom } f = \mathbb{R}_+^2$ and $U = \mathbb{R}_{++}^2$. Since $(-1, 0) \notin \text{dom } f$, we have $\text{dom } f \neq X$.

On U the gradient is $\nabla f(x) = (\ln x_1, \ln x_2)$, and the Hessian $\text{diag}(1/x_1, 1/x_2)$ is positive definite. If $x^k \in U$ converges to a boundary point of $\text{dom } f$ then some coordinate tends to 0, so the corresponding component of $\nabla f(x^k)$ tends to $-\infty$ and $\|\nabla f(x^k)\| \rightarrow \infty$; thus f is essentially smooth.

No boundary point of $\text{dom } f$ carries a subgradient. Indeed, suppose $x_j = 0$ and $p \in \partial f(x)$. Perturbing only the j -th coordinate to $s > 0$ in the subgradient inequality gives $s \ln s - s \geq p_j s$, hence $\ln s - 1 \geq p_j$ for every $s > 0$; taking $s = e^{p_j}$ yields $p_j - 1 \geq p_j$, a contradiction. Therefore $\text{dom } \partial f = U$, and since f is strictly convex on U it is essentially strictly convex. So f is Legendre.

For the conjugate, fix $u \in \mathbb{R}$ and consider $s \mapsto us - g(s)$ on $[0, \infty[$. Its derivative on $]0, \infty[$ is $u - \ln s$ and its second derivative is $-1/s < 0$, so it is maximised uniquely at $s = e^u$, with value e^u ; the boundary value at $s = 0$ is $0 < e^u$. Hence $g^*(u) = e^u$, and by separability $f^*(u) = e^{u_1} + e^{u_2}$, so $\text{dom } f^* = U^* = \mathbb{R}^2$. Finally $x \mapsto (\ln x_1, \ln x_2)$ maps $\mathbb{R}_{++}^2$ bijectively onto $\mathbb{R}^2$, with the coordinatewise exponential as inverse. $\square$

Lemma 2.2. *C is nonempty and compact, $C \subseteq U$, the map c is injective, and*

$$C^* = \{(t, -t^2) : t \in [a, b]\}, \quad \text{cl } C^* = C^* \subseteq U^*.$$

Proof. The map c is continuous on the compact interval $[a, b]$, so C is compact, hence closed, and nonempty. Both coordinates of $c(t)$ are strictly positive, so $C \subseteq \mathbb{R}_{++}^2 = U$. Injectivity holds because $t \mapsto e^t$ is strictly increasing. By Lemma 2.1, $\nabla f(c(t)) = (\ln e^t, \ln e^{-t^2}) = (t, -t^2)$, so C^* is the stated arc; it is a continuous image of $[a, b]$, hence compact and closed, and $U^* = \mathbb{R}^2$ contains it. $\square$

The next lemma is the computation that makes the example tractable: the right Bregman projection onto C decouples into a one-variable problem, and the part depending on x is an additive constant.

Lemma 2.3. *For every $x \in U$ and every $t \in [a, b]$,*

$$D_f(x, c(t)) = f(x) + h_x(t), \quad \text{where} \quad h_x(t) = e^t + e^{-t^2} - x_1 t + x_2 t^2.$$

Proof. For $x, y \in U$, expanding coordinatewise gives the familiar form

$$D_f(x, y) = \sum_{j=1}^2 \left[x_j \ln x_j - x_j - (y_j \ln y_j - y_j) - \ln(y_j)(x_j - y_j) \right] = \sum_{j=1}^2 \left[x_j \ln \frac{x_j}{y_j} - x_j + y_j \right].$$

Substituting $y = c(t) = (e^t, e^{-t^2})$ and using $\ln(x_1/e^t) = \ln x_1 - t$ and $\ln(x_2/e^{-t^2}) = \ln x_2 + t^2$,

$$D_f(x, c(t)) = (x_1 \ln x_1 - x_1 + x_2 \ln x_2 - x_2) + (e^t + e^{-t^2} - x_1 t + x_2 t^2),$$

and the first bracket is exactly $f(x)$. $\square$

Lemma 2.4. *For every $x \in U$ the function h_x is strictly convex on $[a, b]$. Indeed*

$$h_x''(t) = e^t + (4t^2 - 2)e^{-t^2} + 2x_2 \geq e^a + 2x_2 > 0 \quad (t \in [a, b]).$$

Proof. Differentiating h_x twice gives

$$h_x'(t) = e^t - 2te^{-t^2} - x_1 + 2x_2 t, \quad h_x''(t) = e^t + (4t^2 - 2)e^{-t^2} + 2x_2.$$

Since $t \geq a \geq 1/\sqrt{2}$ we have $4t^2 - 2 \geq 0$, so the middle term is nonnegative; and $e^t \geq e^a$ while $x_2 > 0$ because $x \in U$. The displayed bound follows, and it is strictly positive. $\square$

Remark 2.5. Only positivity is used below. On $[1, 2]$ the infimum of $t \mapsto e^t + (4t^2 - 2)e^{-t^2}$ is attained at $t = 1$ and equals $e + 2/e = 3.45404\dots$, so that $h_x'' > e + 2/e$ there, uniformly in $x \in U$. Establishing this sharper bound requires a monotonicity argument; Lemma 2.4 does not.

Lemma 2.6. *C is right D_f -Chebyshev: $\overrightarrow{\Pi}_C(x)$ is a singleton for every $x \in U$.*

Proof. Fix $x \in U$. By Lemma 2.3 the term $f(x)$ does not depend on t , so minimising $D_f(x, \cdot)$ over C is the same as minimising h_x over $[a, b]$; and by injectivity of c (Lemma 2.2) a unique minimising parameter yields a unique minimising point of C . The function h_x is continuous on the compact interval $[a, b]$, so a minimiser exists. If $s \neq t$ were both minimisers then strict convexity (Lemma 2.4) would give

$$h_x\left(\frac{s+t}{2}\right) < \frac{1}{2}(h_x(s) + h_x(t)) = \min_{[a,b]} h_x,$$

which is absurd. Hence the minimiser is unique. $\square$

Remark 2.7. The minimiser need not be interior, and no closed form for it is needed. Since h_x' is strictly increasing by Lemma 2.4, exactly one of three cases occurs: if $h_x'(a) \geq 0$ the minimiser is $t = a$; if $h_x'(b) \leq 0$ it is $t = b$; otherwise $h_x'(a) < 0 < h_x'(b)$ and the minimiser is the unique zero of h_x' in $]a, b[$. These cases are exhaustive because $h_x'(a) < h_x'(b)$. Uniqueness in Lemma 2.6 is insensitive to which case obtains.

Lemma 2.8. *C^* is not convex.*

Proof. The points $(a, -a^2)$ and $(b, -b^2)$ lie in C^* , and their midpoint is $m = (\frac{a+b}{2}, -\frac{a^2+b^2}{2})$. If m belonged to C^* its first coordinate would force the parameter $t = (a+b)/2$, and its second coordinate would then have to equal $-(\frac{a+b}{2})^2$. But

$$\frac{a^2 + b^2}{2} - \left(\frac{a+b}{2}\right)^2 = \frac{(a-b)^2}{4} > 0,$$

so $-\frac{a^2+b^2}{2} < -(\frac{a+b}{2})^2$ and $m \notin C^*$. $\square$

Proof of Theorem 1.3. Lemma 2.1 gives that f is Legendre with $\text{dom } f = \mathbb{R}_+^2 \neq X$, so hypothesis (a) fails. Item (i) is Lemma 2.2, item (ii) is Lemma 2.2, item (iii) is Lemma 2.6, and item (iv) is Lemma 2.8. Since (b) and (c) hold while the conclusion of Fact 1.1 fails, hypothesis (a) cannot be deleted. $\square$

Remark 2.9. We collect the hypotheses and their status. Here X is finite dimensional, f is Legendre, C is nonempty, compact and contained in U , $\text{cl } C^* \subseteq U^*$, and C is right D_f -Chebyshev on all of U . The only failure is the intended one, $\text{dom } f \neq X$. If one reads Fact 1.1 under an additional standing assumption of 1-coercivity, the example still complies: $g \geq -1$ on $[0, \infty[$, so with $M = \max\{x_1, x_2\}$ we have $f(x) \geq M \ln M - M - 1$ and $\|x\|_2 \leq \sqrt{2}M$ for $x \in \text{dom } f$, whence $f(x)/\|x\|_2 \rightarrow +\infty$; outside $\text{dom } f$ the function is $+\infty$. One should not, however, read a standing assumption of supercoercivity for f^* into the setting: for a closed convex f , supercoercivity of f^* is equivalent to $\text{dom } f = X$, which is exactly the hypothesis under test, so requiring it would make Problem 1.2 vacuous.

Remark 2.10. The set C itself is nonconvex, though nothing above requires this; Fact 1.1 does not assume C convex and concludes about C^* . In the variable $r = e^t$, C is the graph of $\psi(r) = e^{-(\ln r)^2}$ over $[e^a, e^b]$, and with $L = \ln r$,

$$\psi''(r) = \frac{e^{-L^2}}{r^2} (4L^2 + 2L - 2),$$

which is strictly positive for $L \geq 1/\sqrt{2}$; so for $a \geq 1/\sqrt{2}$ the function ψ is strictly convex and C is a strictly convex arc, in particular not convex as a set.

3. A CONSEQUENCE FOR A THEOREM OF LUO, MENG, WEN AND YAO

Luo, Meng, Wen and Yao [5] revisit the results surveyed in [3] with the aim of removing coercivity assumptions. Their Theorem 3.12 concerns a right D_f -proximal set C and the statements

- (i) for all $x \in U$ and $y \in C$: $y \in \vec{\Pi}_C(x)$ if and only if $\langle \nabla f(y) - \nabla f(c), y - x \rangle \leq 0$ for every $c \in C$;
- (ii) C is a right D_f -sun, that is, $y \in \vec{\Pi}_C(x)$ implies $y \in \vec{\Pi}_C(\lambda x + (1 - \lambda)y)$ for every $\lambda \geq 0$ with $\lambda x + (1 - \lambda)y \in U$;
- (iv) $\nabla f(C)$ is convex.

They prove that (i) $\Rightarrow$ (ii) with no domain hypothesis; that (i) $\Leftrightarrow$ (ii) if $U = X$; and that (i) $\Leftrightarrow$ (iv) if $\nabla f(U) = U^*$ and f^* is Gâteaux differentiable and strictly convex on U^* . Their Theorem 3.13 states that if f is totally convex at every point of U , if $\nabla f(U) = U^*$, and if f^* is locally uniformly totally convex at every point of U^* , then every boundedly compact right D_f -Chebyshev set is a right D_f -sun.

Our example satisfies (ii) but not (i), and $U \neq X$. We verify both directly rather than by appeal to their theorems, since what is at issue is a configuration their Theorem 3.12(2) excludes.

Lemma 3.1. *Let f and C be as in Theorem 1.3. Then C is a right D_f -sun.*

Proof. Write $y = c(s) \in \vec{\Pi}_C(x)$ for some $x \in U$ and $s \in [a, b]$, and let $\lambda \geq 0$ satisfy $z_\lambda := \lambda x + (1 - \lambda)y \in U$. By Lemma 2.3, the projection of any $w \in U$ is governed by h_w , and by Lemma 2.4 each h_w is strictly convex on $[a, b]$, so h'_w is strictly increasing.

The function

$$h'_w(t) = e^t - 2te^{-t^2} - w_1 + 2w_2t$$

is affine in w , and $h'_y(s) = 0$, since substituting $w = y = c(s) = (e^s, e^{-s^2})$ gives

$$h'_y(s) = e^s - 2se^{-s^2} - e^s + 2e^{-s^2}s = 0.$$

Hence, by affinity,

$$h'_{z_\lambda}(s) = \lambda h'_x(s) + (1 - \lambda)h'_y(s) = \lambda h'_x(s).$$

Now use the three cases of Remark 2.7 for x . If $s \in]a, b[$ then $h'_x(s) = 0$, so $h'_{z_\lambda}(s) = 0$ and s is the unique minimiser of h_{z_λ} . If $s = a$ then $h'_x(a) \geq 0$, so $h'_{z_\lambda}(a) = \lambda h'_x(a) \geq 0$ since $\lambda \geq 0$, and as h'_{z_λ} is increasing the unique minimiser is a . If $s = b$ then $h'_x(b) \leq 0$, so $h'_{z_\lambda}(b) \leq 0$ and the unique minimiser is b . In all three cases $\vec{\Pi}_C(z_\lambda) = \{c(s)\} = \{y\}$. $\square$

Lemma 3.2. *Let f and C be as in Theorem 1.3, put $s = (a + b)/2$ and*

$$x = (e^s - se^{-s^2}, \frac{1}{2}e^{-s^2}).$$

Then $x \in U$ and $y := c(s) \in \vec{\Pi}_C(x)$, but for every $t \in [a, b]$ with $t \neq s$,

$$\langle \nabla f(y) - \nabla f(c(t)), y - x \rangle = (s - t)^2 \frac{1}{2}e^{-s^2} > 0.$$

Hence statement (i) fails.

Proof. Both coordinates of x are positive, since $e^s > se^{-s^2}$ for $s \geq 1/\sqrt{2}$, so $x \in U$. With $x_2 = \frac{1}{2}e^{-s^2}$ we get $2s(e^{-s^2} - x_2) = se^{-s^2} = e^s - x_1$, which rearranges to $h'_x(s) = e^s - 2se^{-s^2} - x_1 + 2x_2s = 0$. As $s \in]a, b[$ and h_x is strictly convex, s is the unique minimiser, so $y = c(s)$ is the right projection of x .

By Lemma 2.1, $\nabla f(c(r)) = (r, -r^2)$, so

$$\nabla f(y) - \nabla f(c(t)) = (s - t, t^2 - s^2) = (s - t)(1, -(s + t)),$$

while $y - x = (e^s - x_1, e^{-s^2} - x_2)$. Therefore

$$\langle \nabla f(y) - \nabla f(c(t)), y - x \rangle = (s - t) \left[(e^s - x_1) - (s + t)(e^{-s^2} - x_2) \right].$$

Substituting $e^s - x_1 = 2s(e^{-s^2} - x_2)$ from the stationarity relation above, the bracket becomes $(2s - s - t)(e^{-s^2} - x_2) = (s - t)(e^{-s^2} - x_2)$, so the whole expression equals $(s - t)^2(e^{-s^2} - x_2) = (s - t)^2 \cdot \frac{1}{2}e^{-s^2}$, which is strictly positive for $t \neq s$. Since $a < b$ such a t exists in $[a, b]$. Statement (i) requires this quantity to be ≤ 0 for every $c \in C$, so (i) fails. $\square$

Corollary 3.3. *Let f and C be as in Theorem 1.3. Then (ii) holds, (i) and (iv) fail, and $U \neq X$. Consequently the hypothesis $U = X$ in [5, Theorem 3.12(2)] cannot be deleted without replacement: the implication (ii) $\Rightarrow$ (i) is false without it.*

Proof. Statement (ii) is Lemma 3.1, statement (i) fails by Lemma 3.2, and (iv) fails by Lemma 2.8. Finally $U = \mathbb{R}_{++}^2 \neq \mathbb{R}^2 = X$. $\square$

Corollary 3.3 says that $U = X$ cannot be struck from Theorem 3.12(2); it does not say that $U = X$ is the weakest hypothesis that would serve there, and some weaker condition on the rays z_λ may well suffice.

Remark 3.4. The proof of (ii) $\Rightarrow$ (i) in [5, Theorem 3.12(2)] applies the sun property along z_λ for arbitrarily large λ , divides by λ , and lets $\lambda \rightarrow \infty$. For the point x of Lemma 3.2 that limit is unavailable: writing $\eta = e^{-s^2}$, we have $(z_\lambda)_2 = \lambda \frac{\eta}{2} + (1 - \lambda)\eta = \eta(1 - \frac{\lambda}{2})$, so $z_\lambda \in U$ exactly for $0 \leq \lambda < 2$. What $U = X$ supplies in that proof is the unbounded reach of the ray, not the sun property.

Remark 3.5. No result of [5] is contradicted. Their Theorem 3.12(1) gives (i) $\Rightarrow$ (ii) unconditionally, which is consistent with (ii) holding and (i) failing. Their Theorem 3.12(3) gives (i) $\Leftrightarrow$ (iv) under hypotheses our example does satisfy, and indeed (i) and (iv) fail together here, as Lemmas 2.8 and 3.2 confirm independently. Their Theorem 3.13 also applies to our example, once one checks that negative entropy is totally convex on U and that f^* is locally uniformly totally convex on U^* , and yields Lemma 3.1 again. Their Corollary 3.14, which concludes convexity of $\nabla f(C)$ from bounded compactness and right Chebyshevness, assumes total convexity at every point of X and $\nabla f(X) = U^*$; both presuppose $U = X$ and therefore do not apply here.

Remark 3.6. Both [3, Fact 3.2] and [5, Theorem 3.12(2)] reach convexity of $\nabla f(C)$ from a Chebyshev-type hypothesis, and both require full domain at the corresponding step, one by assuming $\text{dom } f = X$ outright and the other by requiring $U = X$. Theorem 1.3 and Corollary 3.3 show that the requirement is not an artefact of either proof.

4. THE REMAINING HYPOTHESIS

Hypothesis (b) of Fact 1.1 can be removed altogether. The following is due to Luo, Meng, Wen and Yao; we record it here because it completes the account of Fact 1.1, and we make no claim to it.

Proposition 4.1 (essentially [5, Theorems 3.12 and 3.13]). *Let f be Legendre on $X = \mathbb{R}^n$ with $\text{dom } f = X$, and let $C \subseteq X$ be nonempty and closed. If C is right D_f -Chebyshev, then C^* is convex. In particular hypothesis (b) of Fact 1.1 is redundant.*

Proof. Since $\text{dom } f = X$ we have $U = X$, and C , being closed in $\mathbb{R}^n$, is boundedly compact. By [7, Theorem 26.5], ∇f is a bijection of $U = X$ onto U^* with inverse ∇f^* , and f^* is Legendre; in particular $\nabla f(U) = U^*$, and f^* is differentiable and strictly convex on U^* .

The total-convexity hypotheses of [5, Theorem 3.13] are automatic here. For $y \in X$ and $t > 0$ the set $\{z \in \text{dom } f = X : \|z - y\| = t\}$ is compact, $D_f(\cdot, y)$ is continuous on it, and $D_f(z, y) > 0$ for $z \neq y$ by strict convexity, so $\nu_f(y, t) > 0$. For f^* , fix $u_0 \in U^*$ and $t > 0$ and choose $\rho > 0$ with $\overline{B}(u_0, \rho) \subseteq U^*$; the joint set $K = \{(u, v) : \|u - u_0\| \leq \rho, \|v - u\| = t\}$ is compact and contains no pair with $v = u$, so $m := \min_K D_{f^*} > 0$ and $\nu_{f^*}(u, t) \geq m$ for every $u \in \overline{B}(u_0, \rho)$, whence $\nu_{f^*}^{\text{loc}}(u_0, t) \geq m > 0$. (A bound for each fixed u would not suffice, since such bounds could degenerate as $u \rightarrow u_0$.)

So [5, Theorem 3.13] gives that C is a right D_f -sun, which is (ii); [5, Theorem 3.12(2)], applicable since $U = X$, yields (i); and [5, Theorem 3.12(3)] makes (i) equivalent to convexity of $\nabla f(C)$. $\square$

Thus hypothesis (a) cannot be deleted, and hypothesis (b) need not have been assumed.

5. OPEN QUESTIONS

- (1) *What replaces full domain?* Theorem 1.3 shows that $\text{dom } f = X$ cannot be deleted, but not what the weakest sufficient hypothesis is. Since the right projection onto C for f is the left projection onto C^* for f^* [5, Lemma 3.2], and the obstruction here lies on the dual side, a condition on how C^* approaches ∂U^* seems the natural candidate.
- (2) *Problem 4 of [3]* asks for a characterisation of the right D_f -Chebyshev sets for the negative entropy, and remains open. A solution would subsume Theorem 1.3, which exhibits one such set and computes its gradient image. The example belongs to a two-parameter family, one set for each $1/\sqrt{2} \leq a < b$, so such sets are not scarce.
- (3) *Dimension and geometry.* Our example is planar and C is an arc. We do not know whether the failure can be made worse, for instance with C^* of nonempty interior in its convex hull, nor what happens in $\mathbb{R}^n$ for $n \geq 3$.

VERIFICATION

The statements above were also checked numerically. In directed-rounding interval arithmetic at 40 digits, one harness verifies that D_f agrees with its definition and with the generalised Kullback–Leibler form; that the reduction of Lemma 2.3 holds with constant $f(x)$; that h_x'' is bounded below uniformly on $[1, 2]$; that the minimiser of h_x is unique for a spread of $x \in U$ including near-boundary and far-field points; and the nonconvexity witness of Lemma 2.8. A second harness tests Lemma 3.1 against the definition of a sun over 1,038 pairs (x, λ) , with λ ranging past x to the boundary of U and with x chosen in part to project to the interior of the arc; the projection parameter varies by at most 4×10^{-40} . It also confirms the closed form $(s - t)^2(e^{-s^2} - x_2)$ of Lemma 3.2 and its positivity. These checks are finite; the quantifiers are carried by the proofs.

REFERENCES

- [1] H. H. Bauschke, J. M. Borwein, and P. L. Combettes, *Essential smoothness, essential strict convexity, and Legendre functions in Banach spaces*, Commun. Contemp. Math. **3** (2001), 615–647.
- [2] H. H. Bauschke, X. Wang, J. Ye, and X. Yuan, *Bregman distances and Chebyshev sets*, J. Approx. Theory **159** (2009), no. 1, 3–25.
- [3] H. H. Bauschke, M. S. Macklem, and X. Wang, *Chebyshev sets, Klee sets, and Chebyshev centers with respect to Bregman distances: recent results and open problems*, [arXiv:1003.3127](https://arxiv.org/abs/1003.3127) (2010).
- [4] E. Laude, P. Ochs, and D. Cremers, *Bregman proximal mappings and Bregman–Moreau envelopes under relative prox-regularity*, J. Optim. Theory Appl. **184** (2020), 724–761.
- [5] X.-F. Luo, K. Meng, C.-F. Wen, and J.-C. Yao, *Bregman distances without coercive condition: suns, Chebyshev sets and Klee sets*, Optimization **68** (2019), no. 8, 1599–1624.
- [6] A. Themelis and Z. Wang, *On the natural domain of Bregman operators*, [arXiv:2506.00465](https://arxiv.org/abs/2506.00465) (2025).
- [7] R. T. Rockafellar, *Convex Analysis*, Princeton University Press, 1970.